



**HOOKED AND CHILL: AN INVESTIGATION INTO ENHANCING
ENGAGEMENT AND PROGRESSION IN FISHING ADVENTURE
GAMES THROUGH DYNAMIC ENVIRONMENTS AND
EXPLORATION MECHANICS**

MUHAMMAD ADLIHAQ BIN MOHD ASMADI

**BACHELOR'S DEGREE IN INFORMATION AND TECHNOLOGY
(GAME TECHNOLOGY) WITH HONS.**

2025



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MUHAMMAD ADLIHAQ BIN MOHD ASMADI

**A Thesis submitted in fulfillment of the requirements for the degree of
BACHELOR'S DEGREE IN INFORMATION AND TECHNOLOGY
(GAME TECHNOLOGY) WITH HONS.**

Faculty of Information and Communications Technology

UNIVERSITI TEKNIKAL MALAYSIA MELAKA

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BORANG PENGESAHAN STATUS LAPORAN

JUDUL: HOOKED AND CHILL: AN INVESTIGATION INTO ENHANCING ENGAGEMENT AND PROGRESSION IN FISHING ADVENTURE GAMES THROUGH DYNAMIC ENVIRONMENTS AND EXPLORATION MECHANICS

SESI PENGAJIAN: [2024 / 2025]

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ABSTRACT

Historically, fishing adventure games have tended to feature limited advancement mechanisms and frozen gameplay cycles, which over time frequently leads to decreased player involvement. This study explores how exploration-based mechanics and dynamic environmental systems could be used to improve player engagement and progression in this genre. The study uses a mixed-methods approach that includes both user-centred evaluation and design-based experimentation. A prototype game with exploratory gameplay components (e.g., secret zones, equipment upgrades, and narrative pieces) and real-time environmental changes (e.g., weather patterns, day-night cycles) was created. Structured playtesting sessions with specific user groups were used to gather data, and surveys and observational analysis were used to evaluate player reactions. The findings demonstrate that dynamic settings significantly increase immersion and situational awareness, whereas exploration mechanisms foster curiosity, sustained motivation, and a sense of progress. These findings contribute to the field of game design by providing useful insights into how environmental exploration and interactivity may be used to improve player experience in niche genres such as fishing adventure games.

ABSTRAK

Permainan pengembaraan memancing sering dikritik kerana corak permainan yang berulang dan kurang elemen kemajuan yang bermakna, menyebabkan penurunan minat pemain dalam jangka masa panjang. Kajian ini dijalankan untuk menyiasat bagaimana integrasi persekitaran dinamik dan mekanik penerokaan dapat meningkatkan keterlibatan serta perkembangan pemain dalam permainan jenis ini. Sebuah prototaip permainan dibangunkan dengan ciri-ciri seperti perubahan cuaca, kitaran siang dan malam, serta kawasan tersembunyi yang boleh diteroka. Kaedah kajian melibatkan ujian permainan oleh pengguna terpilih, disusuli soal selidik dan pemerhatian bagi mendapatkan maklum balas. Hasil kajian menunjukkan bahawa perubahan persekitaran mampu meningkatkan suasana permainan yang imersif, manakala elemen penerokaan berjaya menarik minat dan mengekalkan motivasi pemain. Kajian ini memberikan sumbangan kepada pembangunan permainan dengan menekankan kepentingan variasi persekitaran dan elemen penerokaan dalam meningkatkan pengalaman pemain.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In recent years, mobile gaming has seen significant growth, with casual and adventure games gaining popularity due to their accessibility and engaging experiences. Among these, fishing games stand out as a relaxing and rewarding genre. However, many traditional fishing games often rely on repetitive mechanics, static environments, and limited player progression, which can reduce long-term engagement.

This Final Year Project introduces a mobile-based fishing adventure game designed to address these limitations by integrating dynamic environmental elements and exploration-based mechanics. The game aims to provide players with a more immersive and meaningful gameplay experience by simulating real-time changes such as weather, time cycles, and aquatic ecosystems. Additionally, players are encouraged to explore various fishing zones, uncover hidden items, upgrade tools, and discover lore-driven narratives.

The project combines elements of adventure and simulation to create a game that is both relaxing and engaging, while also challenging the player to adapt to ever-changing surroundings. Developed for the Android platform using Unreal Engine, the game is designed with mobile accessibility and performance optimization in mind.

This report outlines the conceptualization, design process, system architecture, development stages, and evaluation results of the game. It also discusses how dynamic features and exploratory gameplay can enhance user engagement and contribute to better progression in mobile games.

1.2 Problem Background

Fishing games have long been popular in the casual and simulation game market, particularly on mobile platforms due to their simple mechanics and relaxing gameplay. However, many existing titles within this genre suffer from design limitations that affect long-term player retention and engagement. These games often rely on repetitive actions, static environments, and minimal interaction beyond casting and catching fish. As a result, players may quickly lose interest, as the gameplay offers little variation or meaningful progression.

Moreover, the lack of dynamic features such as changing weather conditions, day-night cycles, and responsive environments results in a flat and predictable gameplay experience. In addition, most fishing games provide limited incentives for exploration or discovery, which further reduces the potential for immersive and rewarding gameplay. This static design approach does not align with modern player expectations, where interactivity, variety, and personal progression are key factors in maintaining interest.

With advancements in mobile hardware and game engines like Unreal Engine, it is now possible to create more sophisticated and immersive experiences on smartphones. However, these capabilities are often underutilized in casual genres like fishing. Therefore, there is a clear gap in the market for a fishing adventure game that integrates **exploration-based mechanics** to enrich the gameplay experience and encourage sustained player involvement.

This project addresses that gap by designing and developing a mobile fishing game prototype that offers an evolving world and exploratory gameplay, aiming to deliver a more engaging, varied, and meaningful experience for players.

1.3 Problem Statement

Most fishing games available on mobile platforms feature repetitive gameplay, static environments, and limited player interaction, which often lead to a lack of long-term engagement. These games typically do not provide meaningful progression systems or opportunities for exploration, resulting in a shallow and predictable user experience. Despite advancements in mobile technology and the availability of powerful game development engines such as Unreal Engine, many fishing games fail to utilize dynamic features that could significantly enhance gameplay.

There is a clear need for a fishing game that incorporates **dynamic environmental systems**—such as real-time weather changes and day-night cycles—and **exploration-based mechanics** to improve user engagement and progression. Without such features, players are less likely to remain motivated or emotionally invested in the game. Therefore, it is essential to explore how these elements can be effectively implemented to create a more immersive and rewarding fishing adventure experience on mobile platforms.

1.4 Research Aim

The aim of this research is to design and develop a mobile fishing adventure game that enhances player engagement and progression by incorporating dynamic environmental systems and exploration-based gameplay mechanics using Unreal Engine.

1.5 Objectives

- To study player engagement and the factors contributing to the longevity of fishing adventure games, focusing on progression systems and exploration mechanics.
- To study player engagement and the factors contributing to the longevity of fishing adventure games, focusing on progression systems and exploration mechanics.
- To evaluate player experience and gameplay effectiveness through user testing, focusing on immersion, progression satisfaction, and replayability.

1.6 Scope

The goal of this project is to create Hooked and Chill, a fishing adventure game that blends simulation and exploration gameplay features and is compatible with both PCs and mobile devices. The following crucial areas are included in the project's scope:

- **Target Market:** Casual players and simulation fans looking for a fun and relaxed gaming experience are the game's main target audience. Players that prefer stress-free games with minimal hurdles, ambient exploration, and a sense of progression without the intensity of competitive gameplay will find it particularly appealing.
- **Platform:** Developed with Unreal Engine, the game is intended for both PC and Android mobile devices. With the help of Unreal Engine's powerful graphics, physics, and environmental features, this multi-platform deployment guarantees broad accessibility while providing a fluid and aesthetically pleasing gaming experience.
- **Gameplay and Genre:** The adventure and simulation genres include Hooked and Chill. Traditional fishing mechanics are combined with dynamic elements including day-night cycles, real-time weather changes, and a developing world in the gameplay. Through progression and exploration-based objectives, players can discover undiscovered fishing areas, come across rare fish, and uncover story aspects.
- **Theme and Environment:** The game's theme and environment are inspired by nature, which encourages quiet and engaging gameplay. The game has themes of mystery and discovery, motivating players to explore hidden regions, gather rare artefacts, and uncover environmental stories that enhance the game world, even though the main experience centres around leisurely fishing activities.

The project is scoped to deliver a working prototype with core functionalities, including fishing mechanics, dynamic environmental systems, and exploration features. Primary playtesting and evaluation will be conducted on the Android platform to assess player engagement and gameplay effectiveness.

1.7 Significance of Research

Both the academic discipline of game creation and the real-world sector of mobile game design can benefit from this research. Hooked and Chill attempts to address common problems in casual game design, like repetitive gameplay and a lack of player development, by investigating the integration of dynamic settings and exploration elements into a fishing adventure game. The project's results advance our knowledge of how player-driven exploration and environmental interactivity can be used to improve casual simulation game engagement.

From the standpoint of game creation, this project shows how Unreal Engine can be used as a tool to create immersive, mobile-friendly experiences. It demonstrates how sophisticated engine elements, such as environmental changes, real-time weather systems, and non-linear exploration, can be modified for mobile platform efficiency without sacrificing gameplay or visual quality.

This study offers useful insights on player behaviour and preferences for relaxation, immersion, and engagement for the gaming business, especially in the casual mobile sector. The game appeals to a specialised but expanding market that searches out emotionally fulfilling and stress-free gaming experiences by fusing soothing mechanics with deep exploration.

Additionally, this study can be used as a guide for upcoming initiatives that seek to close the gap between adventure and casual simulation genres, providing academics and developers with a starting point for investigating hybrid gameplay models that keep players interested in extended periods of time.

1.8 Report Organization

This thesis is divided into six chapters, which are ordered as follows:

- Chapter 1: Introduction

This chapter introduces the background of the project, problem statement, research aim, research objectives, scope, significance, and target audience. It outlines the motivation behind the development of Hooked and Chill and establishes the foundation for the study.

- Chapter 2: Literature Review

This chapter presents a review of existing research, theories, and related works relevant to fishing games, player engagement, dynamic environments, and exploration mechanics. It identifies gaps in current game design and supports the integration of features implemented in this project.

- Chapter 3: Methodology

This chapter discusses the methodology used to develop the project. The Waterfall Model is adopted as the development framework, consisting of sequential phases including requirement analysis, design, implementation, testing, and maintenance. Each phase is described in detail to show the development flow of the game.

- Chapter 4: Implementation

This chapter describes the implementation of the project, including the design of game features, integration of dynamic systems, user interface, gameplay mechanics, and platform deployment using Unreal Engine.

- Chapter 5: Results and Discussion

This chapter presents the results obtained from the project, particularly the outcome of playtesting on the Android platform. It includes user feedback, performance evaluation, and a discussion of how the implemented features meet the objectives and address the problem statement.

- Chapter 6: Conclusion

The final chapter summarizes the entire project, highlighting key findings, limitations, and suggestions for future improvement. It concludes with a reflection on the research aim and overall project achievements.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature and related works relevant to the development of *Hooked and Chill*, a fishing adventure game that focuses on enhancing player engagement and progression through dynamic environments and exploration mechanics. The review is structured to provide a theoretical and practical foundation for understanding the current state of casual and simulation-based games, particularly within the fishing genre.

The chapter begins by exploring the **fundamentals of game engagement**, focusing on what motivates players in casual and simulation games. It then examines the role of **dynamic environmental systems**, such as weather, time cycles, and environmental feedback, in creating immersive gameplay experiences. The discussion further extends to **exploration-based mechanics** and how they contribute to player retention and progression through discovery, narrative, and reward structures.

Additionally, this chapter highlights several **related games and studies**, identifying key features, strengths, and limitations that inform the design choices made in this project. The objective of this literature review is to identify gaps in current game design approaches and to justify the integration of dynamic and exploratory elements in the proposed game.

By understanding prior research and industry practices, this chapter sets the context for the development of *Hooked and Chill*, ensuring that its design is grounded in proven concepts while also contributing original insights to the field of mobile game development.

2.2 Player Engagement in Casual and Simulation Games

Player engagement refers to the emotional, cognitive, and behavioral investment players commit to a game. In casual and simulation games, engagement is often driven by simplicity, accessibility, and relaxing gameplay loops (Sweetser & Wyeth, 2005). However, studies have shown that engagement can diminish rapidly when gameplay becomes repetitive or lacks meaningful progression (Juul, 2010).

A research study by Yee (2006) suggests that exploration, achievement, and social interaction are key motivational factors in gameplay, even in non-competitive genres. This indicates that to maintain long-term engagement, games must provide continuous incentives,

variety, and a sense of discovery. These principles are critical in designing *Hooked and Chill*, where exploration and environmental changes aim to prevent player fatigue.

2.3 Exploration Based Game Mechanics

Exploration in games is not only about spatial movement but also about discovery, narrative progression, and player agency. According to Korhonen et al. (2009), exploration enhances player curiosity and intrinsic motivation, especially when combined with rewards and hidden content.

In games like *Animal Crossing* and *Raft*, exploration is tied closely to collection, progression, and surprise elements. These games succeed by giving players control over their journey while maintaining a relaxing and non-linear structure. Inspired by these practices, *Hooked and Chill* integrates hidden fishing spots, rare fish, and subtle environmental storytelling to reward player curiosity.

2.4 Existing Fishing Games and Industry Gaps

Current fishing games on mobile platforms, such as *Fishing Clash*, *Ace Fishing*, and *Fishing Hook*, emphasize fast gameplay and competitive ranking systems. While popular, these games often lack exploratory elements and environmental variability, resulting in repetitive gameplay loops.

Academic reviews and user feedback from app stores indicate that many players abandon such games due to the lack of depth and immersive features (Google Play Reviews, 2023). Few games in this genre prioritize slow-paced, peaceful gameplay with progression tied to exploration. *Hooked and Chill* aims to fill this gap by offering a relaxing fishing experience complemented by dynamic systems and hidden content.

2.5 Summary of Literature Review

The literature reveals that while casual and simulation games benefit from accessible gameplay, long-term engagement relies heavily on meaningful progression, exploration, and interactivity. Dynamic environmental systems and exploration-based mechanics have been successfully implemented in other genres but are underutilized in fishing games, especially on mobile platforms.

This review highlights the need for a new approach in the design of fishing games—one that integrates natural change, hidden content, and player-driven discovery to create a more

rewarding experience. These insights form the conceptual foundation for *Hooked and Chill*, guiding its development and feature prioritization.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlines the methodology used in the development of the mobile fishing adventure game *Hooked and Chill*. A systematic and structured approach is essential in ensuring the project is completed efficiently and meets its objectives. For this project, the **Waterfall Model** is adopted as the software development methodology due to its linear and sequential nature, which is suitable for projects with well-defined requirements.

The Waterfall Model divides the development process into distinct phases: **requirement analysis, system design, implementation, testing, deployment, and maintenance**. Each phase must be completed before progressing to the next, allowing for thorough documentation and review at each stage.

The choice of this model is based on the clarity of the game's concept and objectives from the beginning of the development process. Since the core gameplay features and system requirements were defined early, the Waterfall Model provides a structured framework to guide the project from planning through to final implementation and testing on the Android platform.

This chapter will explain each phase of the Waterfall Model in detail and demonstrate how it is applied to the design and development of *Hooked and Chill* using Unreal Engine.

3.2 Methodology

3.2.1 Waterfall Model Overview

The Waterfall Model is a traditional software development methodology characterized by distinct, non-overlapping phases. Each phase must be completed before moving on to the next. This approach provides a systematic framework for project execution and ensures that all aspects of the game's design and development are thoroughly documented and reviewed. The key phases in this methodology are as follows:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing

5. Deployment

6. Maintenance

In the context of *Hooked and Chill*, this approach ensures a focused and methodical development process, from conceptualization through to final implementation

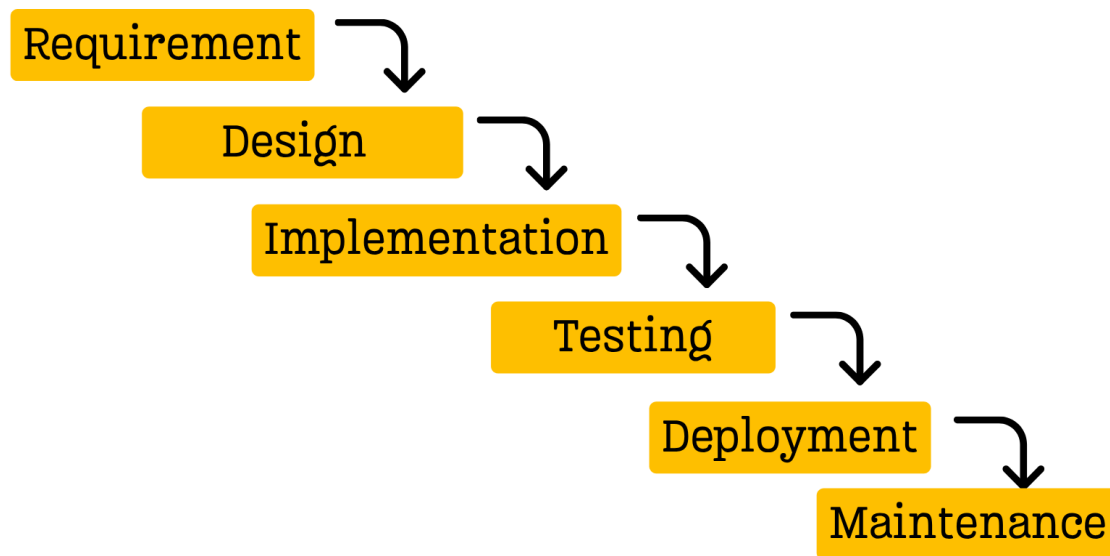


Figure 3.1

3.2.2 Requirement Analysis

Objective:

The primary goal of the requirement analysis phase is to gather, define, and document all the necessary requirements for the game. This phase sets the foundation for the entire development process by ensuring that the project scope, gameplay features, and technical requirements are clear and agreed upon.

Activities:

- User and System Requirements Gathering:
 - Interview stakeholders (e.g., players, testers, game designers) to understand user expectations for the game.
 - Identify key features for player engagement, such as dynamic environmental changes (e.g., weather patterns, day-night cycles), exploration mechanics, and progression systems (e.g., fishing tool upgrades).

- Functional Requirements:
 - Define the gameplay mechanics (interactive fishing system, reward progression, exploration-driven game objectives).
 - Identify the platform for deployment (Android) and ensure that the game's technical requirements are aligned with the capabilities of Unreal Engine 5.
- Non-functional Requirements:
 - Determine performance metrics, such as smooth gameplay with real-time environmental changes, responsiveness to player actions, and stable performance on mobile devices.
 - Define constraints such as battery consumption, mobile platform optimization, and user interface design for mobile devices.

Outcome:

- A comprehensive **requirement specification document** that details the features, gameplay mechanics, performance expectations, and technical platform specifications.

3.2.3 System Design

Objective:

The system design phase takes the requirements and begins transforming them into a structured blueprint. This phase focuses on high-level game architecture, the technical framework, and detailed design elements for each aspect of the game.

Activities:

- High-Level Game Architecture Design:
 - Create a diagram of the overall system, identifying key components such as the **game engine, user interface (UI), input/output mechanisms, and data flow.**
 - Define the interaction between the main game components, such as the environmental systems (weather, day-night cycle) and the fishing mechanics.

- User Interface (UI) Design:
 - Design the game's UI elements, such as the main menu, HUD (heads-up display), inventory management, and player feedback (e.g., in-game notifications, progress bars).
 - Focus on making the interface intuitive and user-friendly, especially for mobile platforms.
- Technical Design:
 - Choose **Unreal Engine 5** as the development platform and ensure it can support the dynamic gameplay features and environmental systems.
 - Develop detailed plans for **level design**, including fishing zones, hidden spots, and the integration of dynamic weather and time cycles.
- Game Mechanics Design:
 - Design the **fishing system**, ensuring realistic fish behavior (e.g., attraction to bait, response to environmental changes).
 - Define the **progression system**, such as unlocking new tools, upgrading fishing gear, and discovering new fishing locations.

Outcome:

- A **detailed system design document** containing architectural diagrams, UI wireframes, and technical specifications for implementing the game.

3.2.4 Implementation

The Implementation phase involved the actual **development of game features**, coding of systems, and integration of designed assets into a working prototype using Unreal Engine.

Activities:

- Programmed the **core gameplay loop**, including casting, reeling, and catching mechanics with real-time tension control.
- Integrated **dynamic environmental systems** such as:
 - Time-based transitions (day to night)

- Weather cycles affecting fish behavior
- Developed **modular levels** representing different fishing zones with hidden locations and rare fish species.
- Implemented an **upgrade system**, where players collect in-game rewards to improve their tools and access new areas.

Outcome:

A playable prototype of *Hooked and Chill* was successfully developed, incorporating dynamic, exploratory, and progression-based gameplay components in a mobile-optimized environment.

3.2.5 Testing

The Testing phase focused on **validating system functionality** and ensuring that all features met the design and performance requirements.

Activities:

- Conducted **functional testing** to confirm:
 - Fishing mechanics and environmental responses worked as intended
 - No major bugs, crashes, or UI inconsistencies were present
- Carried out **user testing sessions** with selected players to gather:
 - Feedback on **immersion, enjoyment, and ease of use**
 - Suggestions for enhancing challenge balance and pacing
- Utilized structured surveys and observational feedback to evaluate:
 - Engagement with exploratory content
 - Effectiveness of reward systems
 - Visual and auditory feedback performance

Outcome:

Testing revealed high levels of satisfaction with gameplay flow and visual feedback. Minor issues identified were addressed before deployment, enhancing overall game stability and polish.

3.2.6 Deployment

This phase covered the **packaging and release** of the prototype for evaluation purposes.

Activities:

- Compiled the game using **Unreal Engine 5.6 mobile deployment settings**.
- Deployed the APK package onto Android test devices.
- Distributed to internal testers for final evaluation and data collection.

Outcome:

The prototype was successfully deployed to Android platforms, ready for further iteration or presentation during final assessment phases.

3.2.7 Maintenance

Although the game was released as a prototype, limited post-deployment updates were made to refine the user experience.

Activities:

- Addressed user feedback from testing sessions, particularly UI improvements and progression pacing.
- Resolved minor bugs and optimized memory usage on mobile devices.
- Prepared codebase and documentation for future expansion (e.g., additional zones or online features).

Outcome:

The game prototype was finalized and documented for academic submission, with a stable and engaging user experience suitable for showcasing gameplay mechanics and system design achievements.

3.3 Conclusion

In conclusion, this chapter has detailed the structured development methodology adopted for the creation of the *Hooked and Chill* game prototype. By implementing the **Waterfall Model**, each phase of the software development lifecycle—ranging from requirement analysis to maintenance—was approached in a clear, systematic manner. This model provided a logical progression that ensured comprehensive planning, robust design, disciplined implementation, and effective evaluation of the game features.

The methodology allowed for the thorough integration of dynamic environmental systems, exploration-based mechanics, and player progression elements—all of which are central to the objectives of this project. Moreover, the methodical approach ensured that the game met both functional and non-functional requirements, particularly with regard to performance on mobile platforms and user engagement.

This foundation supports the next phase of the project, which involves a deeper dive into the technical implementation and results of user testing. The subsequent chapter will focus on how these design and development efforts were realized in practice, highlighting the implementation process and key gameplay outcomes.

Chapter 4: Implementation

4.1 Introduction

This chapter provides a comprehensive overview of the technical implementation of *Hooked and Chill*, a mobile-based fishing adventure game developed using Unreal Engine. The goal of the implementation phase was to transform the system design and documented requirements into a functional prototype. This involved configuring the development environment, integrating game mechanics, designing user interfaces, implementing dynamic environmental systems, and ensuring performance optimization for Android devices. Each component was developed in alignment with the project's objectives to ensure an immersive and engaging user experience.

4.2 Development Environment

The development of *Hooked and Chill* was carried out using Unreal Engine 5.6, selected for its robust features such as real-time rendering, Blueprint visual scripting, and mobile platform support. Unreal Engine's modular system allowed for efficient implementation of gameplay mechanics and dynamic environmental features.

The target platform for the game was Android, due to its wide user base and accessibility. The Android Software Development Kit (SDK) and Android Debug Bridge (ADB) were integrated within Unreal Engine to facilitate mobile testing and deployment.

Hardware used during development included:

- A Windows-based PC equipped with 16GB RAM, AMD Ryzen processor, and NVIDIA GTX 1650 GPU.
- An Android smartphone (mid-range) with at least 4GB RAM for testing and performance profiling.

Additional tools used:

- **Mixamo:** Utilized for rigging and animating character models. Animations such as idle poses, fishing actions, and environmental interactions were sourced from Mixamo and integrated into Unreal Engine.
- **Diversion:** A tool used for managing scene elements, environmental layers, and world-building assets efficiently during level design stages.

- **FAB (formerly Quixel Bridge/Fab Assets):** Used to import high-quality, optimized 3D assets such as terrain, fishing props, environmental foliage, and water shaders. FAB's extensive asset library helped maintain visual consistency while reducing custom modeling time.

This environment provided a robust foundation for asset creation, gameplay logic, and performance optimization across development and testing phases.

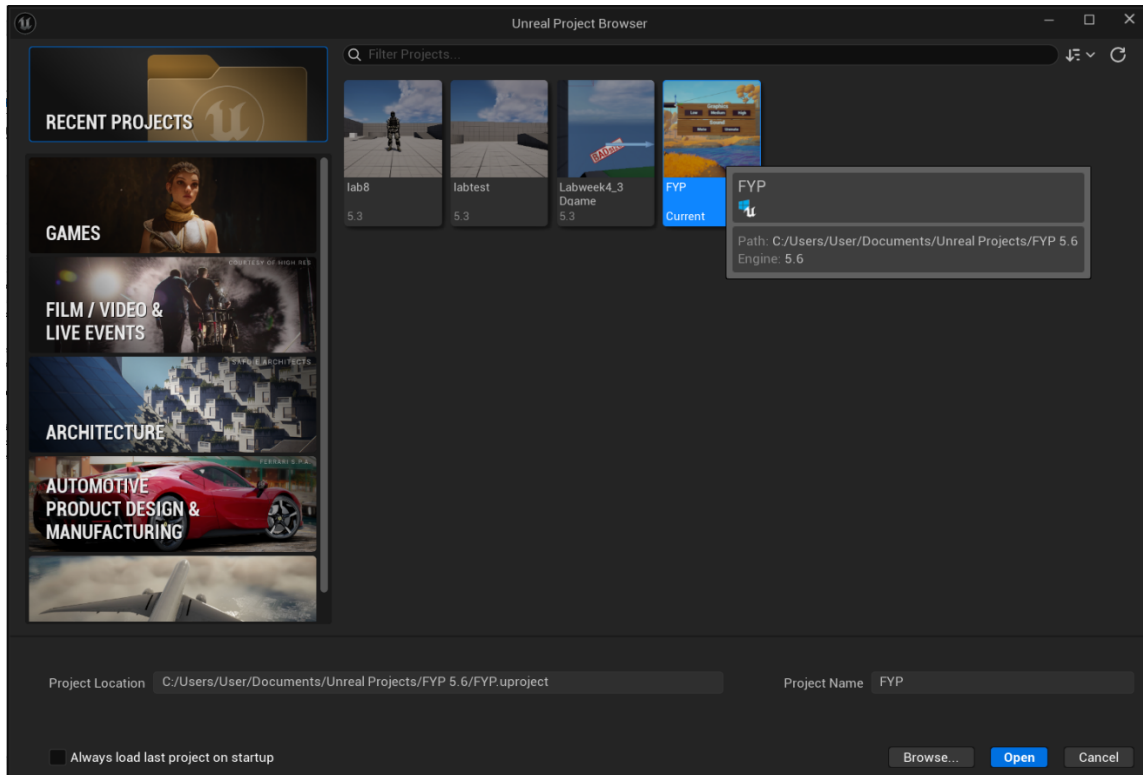


Figure 4.1 Unreal Engine 5.6

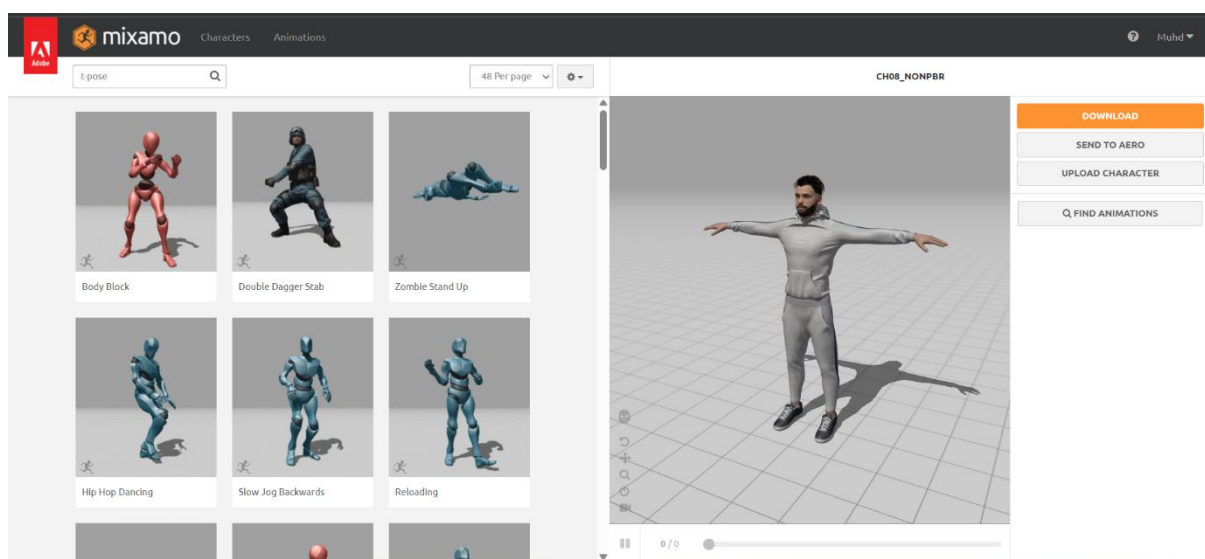


Figure 4.2 Mixamo

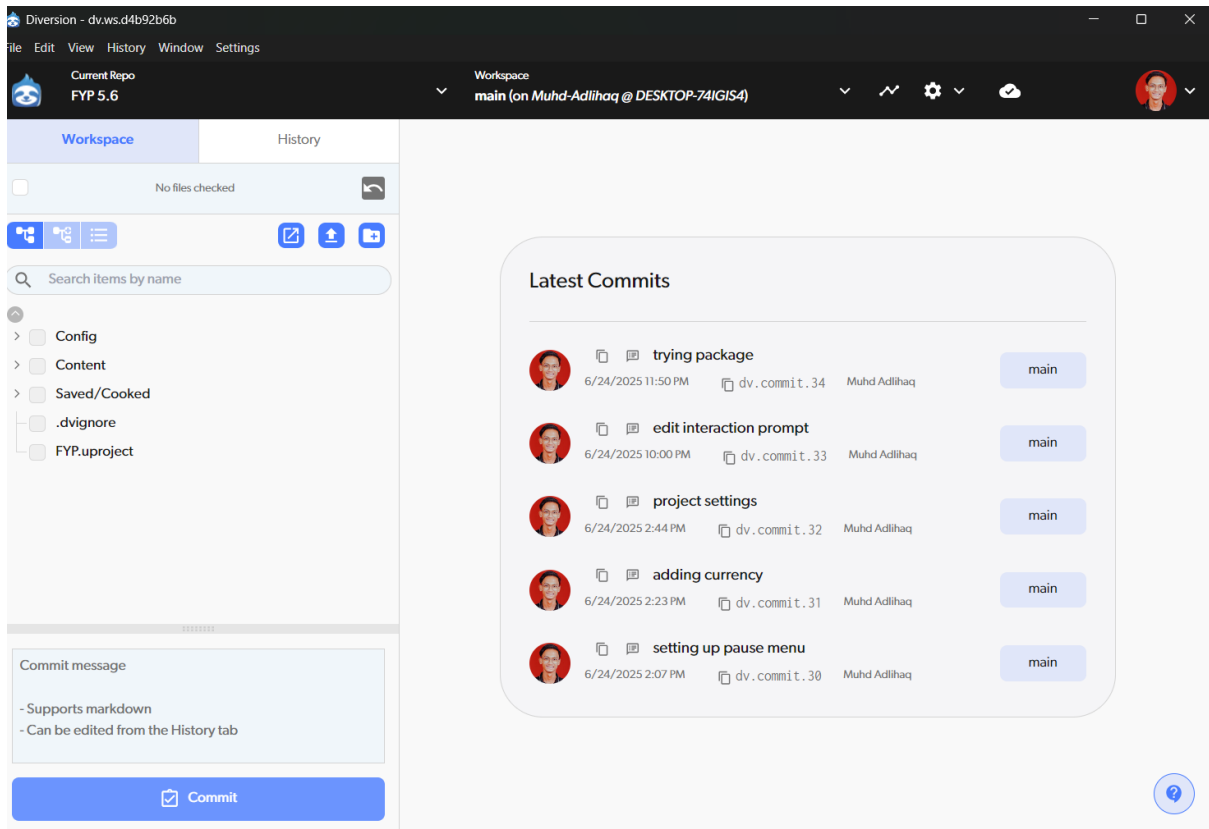


Figure 4.3 Diversion

4.3 Game Architecture Implementation

The game architecture was structured into several core modules to facilitate modular development and future scalability. The primary systems implemented included:

- **Fishing System:** This core mechanic simulates casting, baiting, tension control, and catching fish. The implementation used Unreal Engine's Blueprint system, where touch input controls the cast distance and reeling tension. A real-time "tension meter" determines catch success based on player skill and timing.
- **Progression System:** Players earn in-game currency by catching fish, which can be used to unlock new tools and fishing zones. This system was implemented using a simple data-driven approach, where player actions update variables stored in the save system.
- **Environment System:** A timeline-based system controls weather changes and day-night cycles. These affect fish behavior and availability, encouraging strategic

gameplay. Unreal's Environment Lighting and Sky Sphere features were customized to reflect real-time transitions.

- **Exploration Features:** Hidden fishing spots were created by placing trigger volumes in level design. Players can discover these areas by exploring the map, unlocking rare fish or story-based lore items.

All components interact through the game loop controller, which manages player input, updates the UI, and coordinates in-game events.

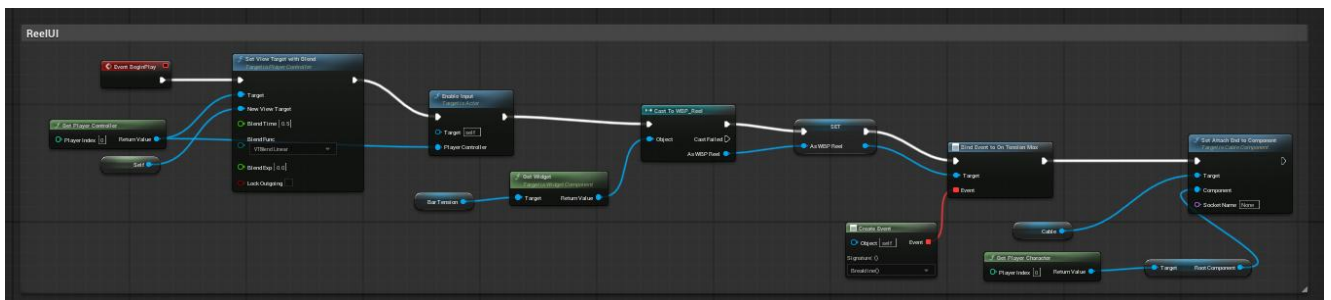


Figure 4.3.1 Reel Tension UI

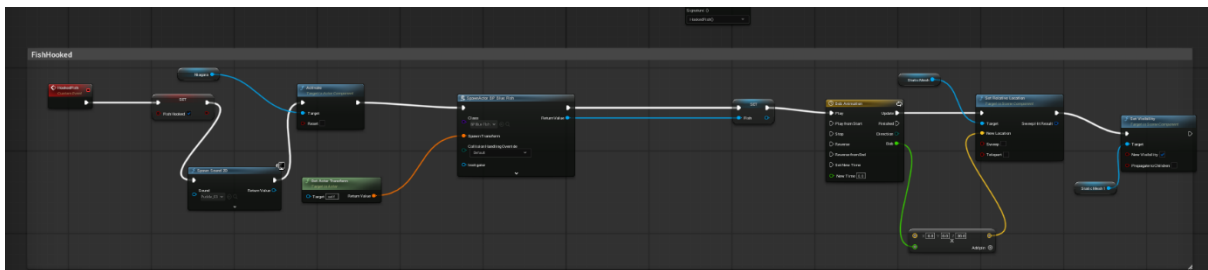


Figure 4.3.2 Fish Hooked

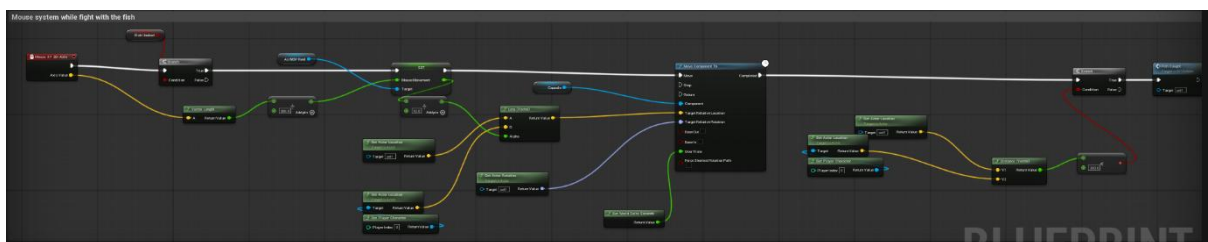


Figure 4.3.3 Mouse acceleration to fight with the fish and balance the tension reel

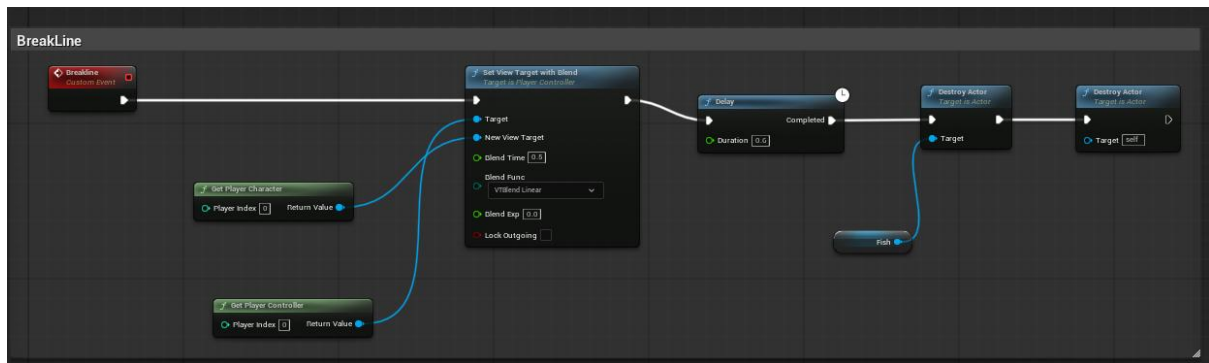


Figure 4.3.4 When fishing line breaking

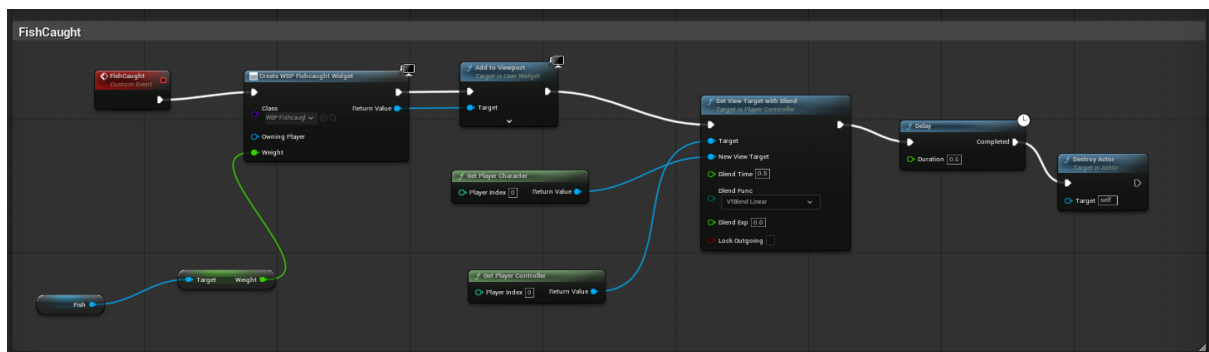


Figure 4.3.5 Fish Caught

4.4 User Interface Implementation

User experience (UX) was a major focus in *Hooked and Chill*. The interface was designed to be minimalist, touch-friendly, and context-sensitive to avoid screen clutter on mobile devices.

- **Main Menu:** Features options to Start Game, Access Settings, and Exit. Designed with clear buttons and scalable font sizes for various screen resolutions.
- **In-Game HUD:**
 - Displays current time of day and weather.
 - Shows the fishing tension meter during gameplay.
 - Provides visual feedback such as fish proximity, casting range, and reeling tension.
- **Inventory & Upgrade Menu:** Allows players to view collected fish, check tool stats, and use in-game currency to upgrade their rod or unlock new areas. Implemented using Unreal's UMG system.
- **Touch Interaction:** Swipe and tap gestures are mapped to reeling and casting actions, providing a tactile feel to the fishing experience.

Each UI element was tested for responsiveness and performance on Android devices to ensure a smooth user experience.



Figure 4.4.1 UI in Game while fishing



Figure 4.4.2 While player wanna take the quest

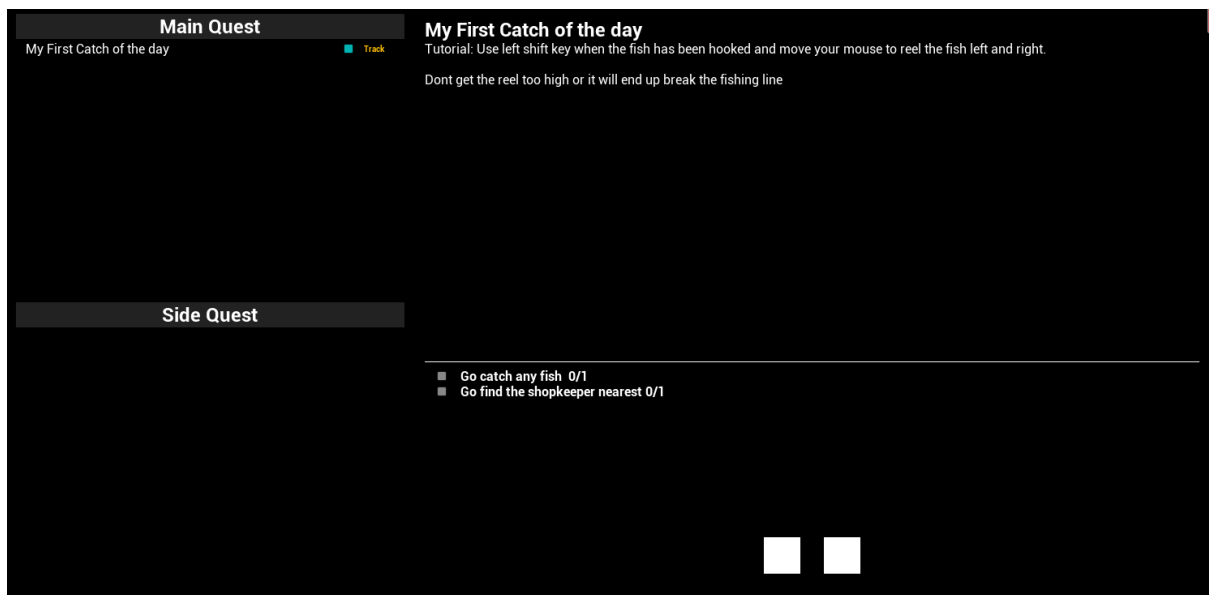


Figure 4.4.3 Quest UI



Figure 4.4.4 Quest Tracker UI

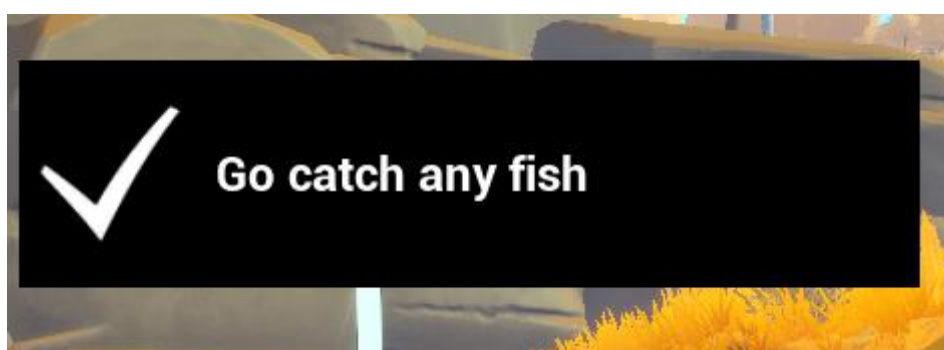


Figure 4.4.5 Quest Completed UI



Figure 4.4.6 Pause Menu



Figure 4.4.7 Shopkeeper Interaction



Figure 4.4.8 Loading Screen

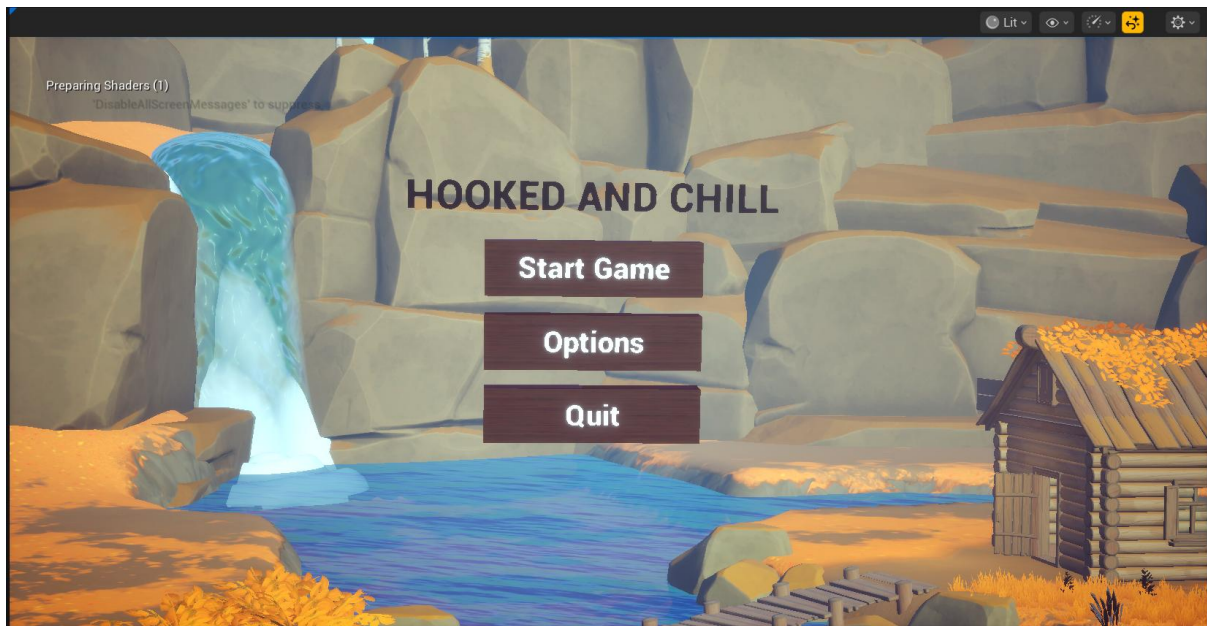


Figure 4.4.9 Main menu

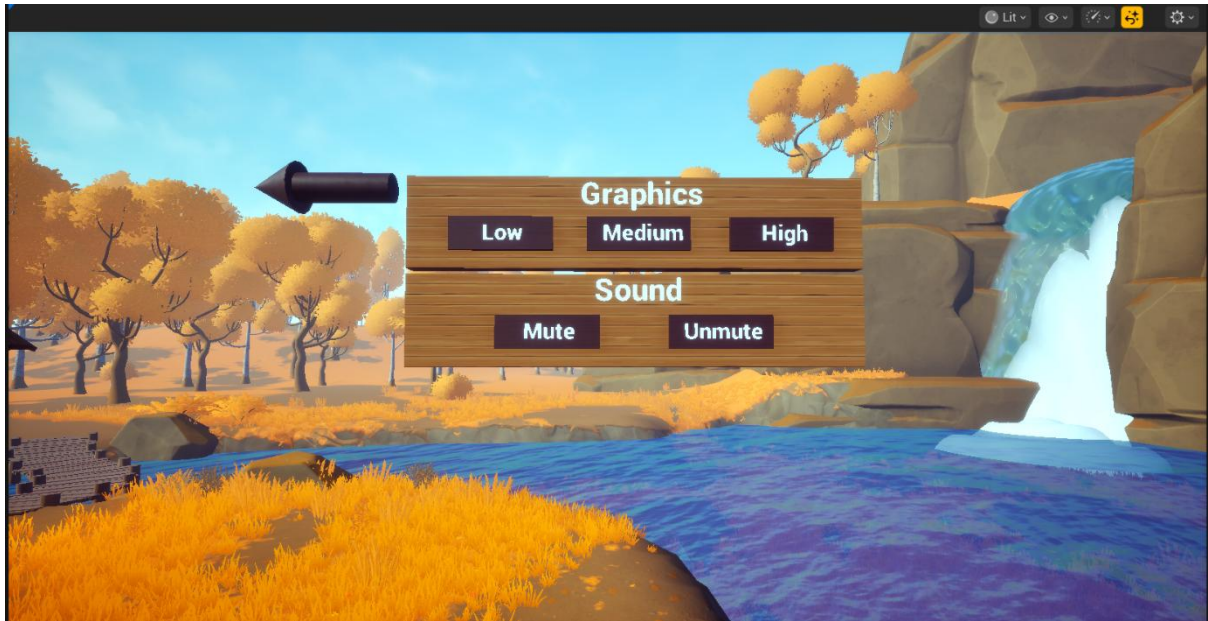


Figure 4.4.10 Options Menu

4.5 Game Logic and Mechanics

Gameplay mechanics in *Hooked and Chill* were implemented to provide not only interactive fishing experiences but also meaningful player progression through a **Quest System**, **fish AI**, **tension control**, and **unlockable content**. These systems work together to create a cohesive and engaging loop of exploration, challenge, and reward.

Fish AI

Basic artificial intelligence was implemented for fish behavior using Unreal Engine's Blueprint system. Fish entities follow patrol paths and react dynamically to bait proximity. Their appearance is influenced by weather, time of day, and zone rarity, encouraging players to explore various locations under different environmental conditions.

Tension Control System

The fishing mechanic includes a real-time tension bar where players must balance reeling speed to maintain optimal pressure. If tension exceeds the threshold, the fish escapes. This system was created using Unreal's Timeline nodes and adds a level of skill-based interaction, simulating realistic fishing difficulty.

Quest System

A **Quest System** was designed to provide narrative depth and structured player objectives. Quests guide players through key gameplay features, reward exploration, and enhance immersion.

- **Types of Quests:**
 - **Main Quests:** Advance the core storyline
 - **Side Quests:** Offer additional challenges and rewards
 - **Exploration Quests:** Encourage discovery of hidden areas or environmental secrets.
- **Implementation:**
 - Quests are stored in a structured Data Table with titles, objectives, conditions, and rewards.

- A **Quest Manager** Blueprint handles activation, progress tracking, and completion logic.
- UI widgets display active quests, objectives, and completion rewards.
- **Rewards:**
 - Completion of quests grants experience points, in-game currency, equipment upgrades, or unlocks new dialogue or lore elements.
 - Certain quests are time-sensitive or dependent on environmental triggers (e.g., weather or time of day).
- **Narrative Integration:**
 - Quest descriptions include environmental storytelling and background lore to make the world feel alive.
 - Example: *“A local fisherman lost a legendary rod at sea during a storm. Recover it from the Hidden Cove to gain his trust.”*

This Quest System adds structure to gameplay while maintaining an open-ended and relaxing pacing. It also gives purpose to the player’s actions beyond fishing and upgrades, tying into the game’s exploration and adventure theme.

Reward System

Players are rewarded based on fish rarity, quest completions, exploration, and skillful fishing techniques. Rewards include in-game currency, tools, and experience points. This system reinforces a sense of progression and accomplishment.

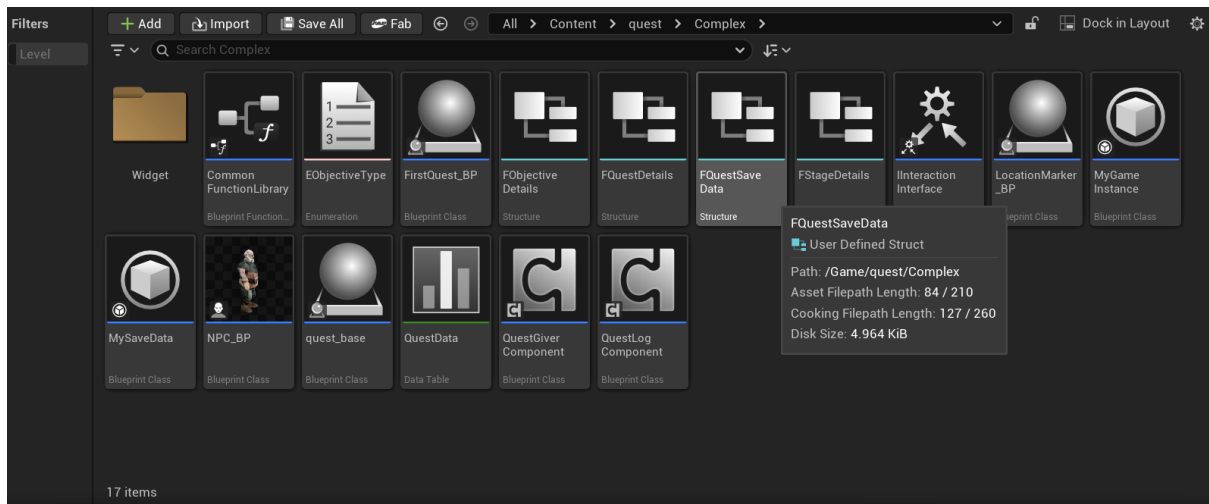


Figure 4.5.1 Quest system

4.6 Dynamic Environment Integration

Environmental realism and variety are key features that distinguish *Hooked and Chill* from traditional fishing games.

- **Day-Night Cycle:** Implemented using Unreal's light direction and time logic, this system transitions smoothly between morning, afternoon, evening, and night. Certain fish only appear at specific times, adding strategic depth.
- **Weather System:** A randomized weather engine controls conditions such as clear skies, light rain, and thunderstorms. These events influence fish behavior and also change visual effects, ambient audio, and water physics.
- **Ambient Audio and Visual Effects:**
 - Background music and ambient sounds change with weather and time.
 - Particle effects (e.g., rain splashes, fog) enhance immersion.
 - Subtle camera shakes and haptic feedback add realism when a fish bites or pulls on the line.

This system brings the world to life and aligns with the project's aim of creating a calm yet dynamic fishing adventure.

4.7 Testing Build and Debugging

Testing and debugging were essential in refining the prototype and ensuring its readiness for evaluation.

- **APK Packaging:** The project was built and deployed as an APK file using Unreal's Android packaging tools. Installation was performed via USB and ADB for direct testing on mobile devices.
- **Performance Testing:**
 - FPS (frames per second) monitoring ensured smooth gameplay on target devices.
 - Heat and battery consumption were monitored to meet non-functional requirements.
- **Bug Fixing:**
 - Issues such as touch input delays, UI scaling problems, and audio overlaps were identified and resolved.
 - Fish AI pathing and reward calculations were adjusted after receiving initial feedback.

Overall, the prototype was stable and met the required performance benchmarks.

4.8 Conclusion

The implementation phase successfully translated the design specifications of *Hooked and Chill* into a playable prototype. Each component—from gameplay mechanics and UI to environment systems and platform deployment—was developed using a modular and iterative approach. The integration of dynamic elements such as weather, time of day, and exploration mechanics significantly enriched the gameplay experience. The prototype was optimized for Android and tested under real-world conditions, validating its technical feasibility and user appeal. This chapter lays the groundwork for the next stage: evaluating the effectiveness of the game through user testing and performance analysis.

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